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CSC 351
HW3

1. Retrieve the plant name, country name, operator name, fuel category, fuel name, capacity (in MW), and commission year for all power plants without using table aliases. Sort the results in descending order by capacity.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'hw3dbs' selected. The central pane shows a SQL query:

```
1. set SQL_SAFE_UPDATES=0;
2. set FOREIGN_KEY_CHECKS=0;
3.
4. #1. Retrieve the plant name, country name, operator name, fuel category, fuel name,
5. #capacity (in MW), and commission year for all power plants without using table aliases.
6. #Sort the results in descending order by capacity.
7. select power_plants.PlantName,
8. power_plants.CapacityMW, power_plants.CommissionYear,
9. countries.CountryName, operators.OperatorName,
10. fuel_types.FuelName, fuel_types.FuelCategory
11. from power_plants
12. inner join countries
13. on power_plants.CountryCode = countries.CountryCode
14. inner join fuel_types
15. on power_plants.FuelID = fuel_types.FuelID
16. inner join operators
17. on power_plants.OperatorID = operators.OperatorID
18. order by CapacityMW desc;
19.
```

 The bottom pane shows the 'Result Grid' with 11 rows of data. The columns are PlantName, CapacityMW, CommissionYear, CountryName, OperatorName, FuelName, and FuelCategory. The data is sorted by CapacityMW in descending order.

PlantName	CapacityMW	CommissionYear	CountryName	OperatorName	FuelName	FuelCategory
Three Gorges	22500	2003	China	China Yangtze Power	Hydro	Renewable
Itaipu	14000	1984	Brazil	Elektrobras	Hydro	Renewable
Tuchuan	13200	1995	China	China Yangtze Power	Coal	Fossil
Kashiwazaki-Kariwa	8212	1985	Japan	Teppo Energy	Uranium	Nuclear
Grand Coulee	6809	1942	United States	Global Hydro Inc	Hydro	Renewable
Vindhyachal	4760	1987	India	NTPC Limited	Coal	Fossil
Sanchevade	3200	1981	Germany	Rhein Power AG	Coal	Fossil
Loy Yang	2210	1984	Australia	Snowy Hydro	Coal	Fossil

2. Retrieve the plant name, country code, calendar year, and annual generation (in GWh) for all power plants for the year 2024 without using table aliases. Sort the results in descending order by generation.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'hw3dbms' selected. The main editor shows a SQL query (lines 19-37) that retrieves plant names, country codes, years, and annual generation for 2024, sorted by generation in descending order. The 'Result Grid' at the bottom shows the query results (lines 19-37) with columns: PlantName, CountryCode, year, and generationwh. The results are as follows:

PlantName	CountryCode	year	generationwh
Three Gorges	CHN	2024	18800
Itaipu	BRA	2024	89000.1
Turkey	CHN	2024	71000.3
Vindhyachal	IND	2024	35600.8
Grand Coulee	USA	2024	21500.5
Janschwalde	DEU	2024	18400.2
Loy Yang	AUS	2024	14200.4
Kashiwazaki-Kariva	JPN	2024	0

- Retrieve the plant name, country code, calendar year, annual generation (in GWh), and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases. Sort the results in ascending order by CO2 emissions.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'hw3dbms' selected. The main editor shows a SQL query (lines 35-53) that retrieves CO2 emissions (in tonnes) for all power plants for the year 2024, sorted by CO2 emissions in ascending order. The 'Result Grid' at the bottom shows the query results (lines 35-53) with columns: co2emissionstons, CountryCode, year, PlantName, and generationwh. The results are as follows:

co2emissionstons	CountryCode	year	PlantName	generationwh
0	CHN	2024	Three Gorges	21500.5
0	JPN	2024	Kashiwazaki-Kariva	21500.5
0	USA	2024	Grand Coulee	71000.3
0	USA	2024	Grand Coulee	21500.5
0	USA	2024	Grand Coulee	14200.4
0	JPN	2024	Kashiwazaki-Kariva	18400.2
0	BRA	2024	Itaipu	18400.2
0	USA	2024	Grand Coulee	18400.2
0	CHN	2024	Three Gorges	35600.8
0	JPN	2024	Kashiwazaki-Kariva	35600.8
0	BRA	2024	Itaipu	35600.8
0	USA	2024	Grand Coulee	35600.8
0	CHN	2024	Three Gorges	89000.1
0	JPN	2024	Kashiwazaki-Kariva	89000.1
0	BRA	2024	Itaipu	89000.1
0	USA	2024	Grand Coulee	89000.1
0	CHN	2024	Three Gorges	0

- Write a SQL query using a Common Table Expression (CTE) to calculate the total cumulative power generation (in GWh) for each operator across all available years

without using table aliases. Retrieve the operator name, headquarters country, and their total generated power, and sort the results in descending order by total generation.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

54
55 with operator_generation as (
56   select power_plants.operatorid,
57   sum(generation_records.generationgwh) as totalgenerationgwh
58   from power_plants
59   join generation_records
60   on power_plants.plantid = generation_records.plantid
61   group by power_plants.operatorid
62 )
63 select
64   operators.operatorname, operators.headquarterscountry,
65   operator_generation.totalgenerationgwh
66   from operator_generation
67   join operators
68   on operator_generation.operatorid = operators.operatorid
69   order by operator_generation.totalgenerationgwh desc;
70
71 #5. Write a SQL query using two separate Common Table Expressions (CTEs)--one to calculate total power generation by
72   country code and another to calculate total CO2 emissions by country code without using table aliases.

```

The Result Grid shows the following data:

operatorname	headquarterscountry	totalgenerationgwh
China Yangtze Power	CHN	169800.3
Elecbras	ISR	89000.1
NTPC Limited	IND	36600.8
Global Hydro Inc	USA	21000.5
Tepco Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepco Energy	JPN	0

- Write a SQL query using two separate Common Table Expressions (CTEs)--one to calculate total power generation by country code and another to calculate total CO2 emissions by country code without using table aliases. Then, join these CTEs with the countries table to display the country name, total generation (in GWh), and total CO2 emissions (in tonnes). Sort the results in descending order by total generation.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

76 with power_country as
77 (
78   select power_plants.CountryCode, sum(generation_records.generationgwh) as gwhsum
79   from power_plants
80   inner join generation_records
81   on power_plants.plantid = generation_records.plantid
82   group by power_plants.CountryCode
83 ),
84 emission_country as
85 (
86   select power_plants.CountryCode, sum(emission_metrics.co2emissionstonnes) as emsum
87   from power_plants
88   inner join emission_metrics
89   on power_plants.plantid = emission_metrics.plantid
90   group by power_plants.CountryCode
91 )
92 select countries.CountryName, emission_country.emsum,
93   power_country.gwhsum
94   from countries
95   inner join power_country
96   on countries.CountryCode = power_country.CountryCode
97   inner join emission_country
98   on countries.CountryCode = emission_country.CountryCode
99   order by power_country.gwhsum desc;
100

```

The Result Grid shows the following data:

CountryName	emsum	gwhsum
China	6200000	169800.3
Brazil	0	89000.1
India	3100000	36600.8
United States	0	21000.5
Germany	2450000	18400.2
Australia	1950000	14200.4
Japan	0	0

6. EER Diagram

